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China Industrials | Asia Pacific

The Humanoid Signal: Progress Check, WRC Preview

Humanoid robots shipment achieved fast growth with early commercialization in 1H26. Use cases are broadening toward real tasks; reliability, generalization, and data remain key constraints.

1H26 humanoid robots sales volume sees fast growth: Global shipments reached 19k units in 1H26, up 272% YoY, with Chinese players accounting for 97%. Although ~65% of shipments remain in entertainment, education/R&D and data collection, use cases are broadening into industrial and commercial applications. By form factor, half-size bipeds remain the largest-volume (~50%), while wheel-based are expanding rapidly (~29%) in commercial and industrial settings. We expect more 1H26 pilots to convert into broader deployments from 2H26, and forecast China humanoid shipments will reach 50k units in 2026e and 446k by 2030e.

Overall industry's key focus is shifting to model, data, application and dexterous hands from body hardware: For models, the paths have yet to converge; we think it's likely that various approaches (VLA, world-model, multimodal model) can coexist for differently purposed training. Similarly for data, various types of data and data collection (ego-centric data, videos, synthetic data, tele-operated robot data, etc.) are heavily explored, and being debated on usefulness and costs for robot training. Data scaling law is becoming consensus for humanoids, and leading Chinese humanoid robot companies expect real physical world data through robot deployment to become a factor in the data flywheel in the next couple of years. For applications, we expect real-world deployments to account for a rising share of shipments as projects move from verification to scaled rollout.

WRC 2026 (Aug 19–23, Beijing) preview: This year, the World Robot Conference (WRC) is in larger scale with 300+ exhibitors (+36% YoY), 2,000+ exhibits (+27% YoY) and 150+ product debuts (+21%). Use cases exploration will be key to watch, along with some new products/model launches. Government and ecosystem support remains strong, including participation from 49 central SOEs, the planned launch of a central SOE robot innovation consortium and a global robot application program. Overall, we expect to see strong public and media attention, supporting sector sentiment.

Stock implications - positive: As deployment gradually starts, the bar for component suppliers should rise from product qualification to reliability, consistency, yield, manufacturing scale and cost, favoring suppliers with proven mass-production capabilities. Sector sentiment has been weak since May 2026, but several catalysts appear ahead, including Unitree's IPO (according to 10 Aug 2026 CNA report, >8,000x over-subscribed on retail end), WRC and the Humanoid Games, potential Tesla Optimus update, etc. We prefer Leaderdrive, Hengli Hydraulic and Shuanghuan.

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Industry View

In-Line

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World Robot Conference 2026 Preview

World Robot Conference 2026 will be held from Aug 19-23; it is the largest global robotics events, per CCTV.

We expect continued strong whole value chain presence, public interest and wide media exposure, supporting sector sentiment.

Scale steps up materially, broader base: According to the [Vice-Chairman of Chinese Institute of Electronics](#) (CIE, the co-host of WRC), this year's event will host 300+ exhibitors (+36% YoY), 2,000+ exhibits (+27%), and 150+ product debuts (+21%) across four exhibition halls (up from three). Beyond humanoid robot integrators, component suppliers, the international collaboration zone, and solutions across manufacturing, F&B/retail, healthcare, elderly care, and emergency rescue, this year adds a consumer-facing interactive services hall.

Sustained government support for an early stage intelligent robot industry: A dedicated delegation of central SOEs (state-owned-enterprise) will participate, with 49 SOEs across 12 application scenarios. Moreover, **SASAC will announce a central SOE robot innovation consortium**, focused on joint technology development, shared application scenarios, and commercialization, which could improve access to real operating environments and accelerate testing, data collection, and deployment. This shows government continued effort to establish early use cases/applications. Earlier in June, [MIIT and SASAC](#) reportedly announced a new initiative that targets 10k-level deployment capacity and >100 high-value applications by end-2026. Local governments and SOEs must identify operational sites (≥ 20 /province, ≥ 10 /central SOE) for humanoid training.

More unified efforts for global ecosystem effort: The event will launch a Global Robot Application Exploration Program, offering mass-produced robots to selected global innovation teams for free trials.

New product launches: WRC is close to the World Artificial Intelligence Conference (held in July 2026 in Shanghai), where we already saw some new products such as Unitree GD01. A few companies are also expected to release new product at WRC. The announced ones include:

- **Brain:** GigaWorld new foundational model and locomotion model, Beijing Humanoid Innovation Center's Unified model - Pelican-Unify
- **Body:** Inspire-Robots tendon-drive dexterous hand, Xynova new direct-driven dexterous hand Prima 1
- **Robot:** Beijing Humanoid Innovation Center's Tiangong Omni, GigaAI new general-purpose bipedal humanoid robot

What Are the Latest Developments, and Where Are the Bottlenecks?

1H26 Shipments - At early commercialization

China accounted for almost all global humanoid shipments in 1H26: According to a [Bloomberg/Smart Analytics Global](#) report (10 Aug 2026), 1H26 global shipments reached 19k, surging 272% from 5k in 1H25, indicating the industry moved from prototype validation toward early-scale commercialization in 1H26. Chinese players contributed 97% of the total. A leading Chinese integrator increased its market share to 44% from 25% in 1H25, while Unitree accounted for 31%, vs. 43% a year earlier. ([Exhibit 1](#)).

Moving toward task applications: Entertainment, reception, education/R&D and data-collection demand still accounted for roughly 65% of shipments ([Exhibit 2](#)), reflecting relatively low deployment barriers. However, industrial orders are increasing across automotive, logistics and other manufacturing applications, while retail, hotels and other interactive commercial use cases are also emerging. In our view, the next stage of commercialization will increasingly depend on reliability, autonomy and economic value in real operating environments rather than locomotion performance alone.

Increasing mix of wheel-based humanoid shipment, per MIR data ([Exhibit 3](#)).

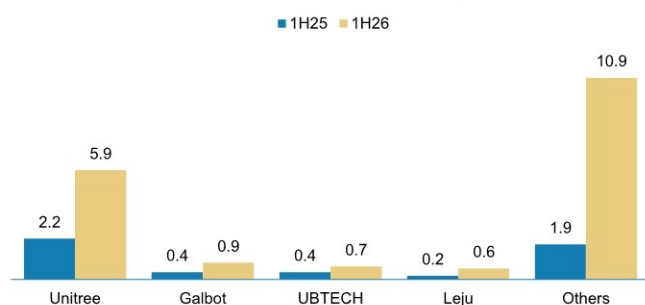
- **Half-sized bipedal humanoids** remain the largest-volume segment, supported by established applications in entertainment, exhibitions and reception, education and R&D demand.
- **Wheel-based humanoids** are seeing the fastest expansion in commercial and industrial deployments because they offer greater stability and provide a practical platform for validating manipulation and embodied-AI capabilities.
- By contrast, **full-size bipedal humanoids** remain more R&D-intensive and are generally produced in smaller batches, with vendors prioritizing hardware/software iteration and longer-term household applications rather than aggressive capacity expansion.

Strong capacity growth: Per MIR, humanoid production capacity growth was supported by a combination of self-built production lines, contract manufacturing and government-backed facilities, with leading players increasingly achieving thousand-unit monthly production and improving supply reliability.

Note: We acknowledge that humanoid robot shipment estimates vary across sources, including Smart Analytics Global and MIR Databank, reflecting the industry's still-early stage and limited disclosure from predominantly private companies. We therefore view these estimates primarily as directional indicators of industry growth and commercialization trends.

Exhibit 1: Per SAG, global humanoid robot shipment increased 272% to 19.1k in 1H26

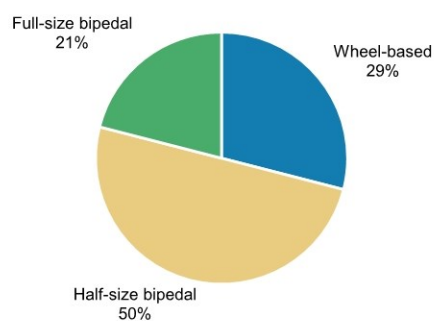
Global Humanoid Shipment ('k unit)



Source: Smart Analytics Global, Morgan Stanley Research.

Exhibit 3: 1H26 China humanoid shipments by type

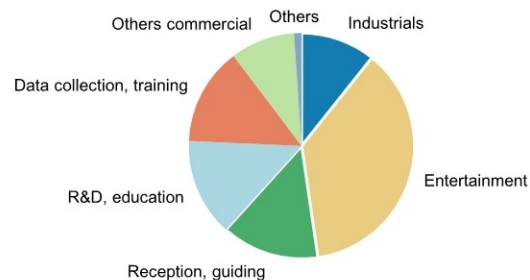
1H26 China Humanoid Shipment by Type



Source: MIR Databank, Morgan Stanley Research

Exhibit 2: We saw more commercial use cases in 1H26

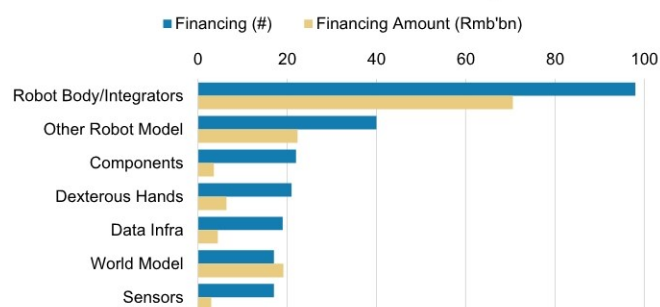
1H26 China Humanoid Shipment by Applications



Source: MIR Databank, Morgan Stanley Research

Exhibit 4: Financing activities are expanding from robot body and components toward robot model (including world model), dexterous hand, data

1H26 Humanoid Financing

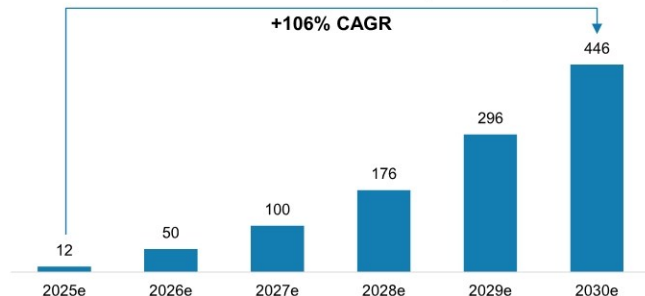


Source: GGII, Morgan Stanley Research.

We expect China humanoid shipments of 50k units in 2026e and 446k units by 2030e, underpinned by integrators' competition to accumulate deployment scale and close the real-world data feedback loop. Use cases are broadening across factory lines, logistics sorting, unmanned retail, and commercial services, etc. We expect 1H26 pilot projects to begin converting into broader deployments from 2H26.

Exhibit 5: We forecast China humanoid shipment will reach 50k/446k in 2026e/2030e...

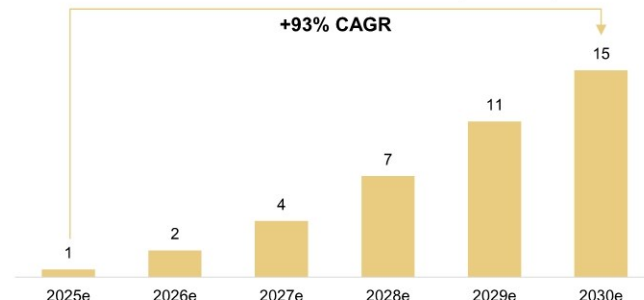
China Humanoid Sales ('k unit)



Source: Morgan Stanley Research (e) estimates

Exhibit 6: ...implying market size of US\$2bn/\$15bn in 2026e/2030e

China Humanoid Market Size (US\$'bn)



Source: Morgan Stanley Research (e) estimates

Hardware: Basic capability being increasingly solved; reliability and higher performance are the next constraints

Reliability and continuous operation remain key gaps before humanoids can replace human labor at scale: Most leading humanoids can now perform basic locomotion and manipulation, but continuous operation remains well below the standard required for large-scale labor substitution. Current deployments still show downtime for charging, cooling, maintenance and debugging. Even during the 2026 E-Town Humanoid Marathon, some robots required mid-race cooling or lubrication despite significant improvements in durability.

Higher reliability requirements raise the bar for component suppliers: As volumes ramp, competition should shift from simply having a qualified product toward capacity ramp-up, precision-processing know-how, product consistency and yield, which ultimately determine reliability and cost. We expect this transition to favor suppliers with proven mass-production capabilities and accelerate consolidation in the currently fragmented component landscape.

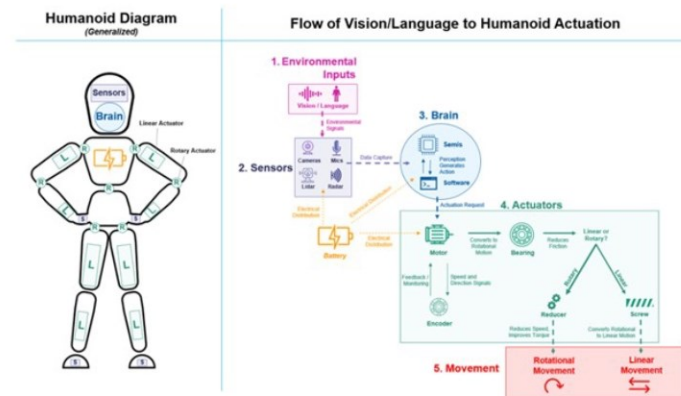
Higher performance is becoming a broader requirement as humanoids move from demonstrations toward productive work: For labor substitution, the question will increasingly shift from whether a robot can complete a task to how much it can carry, how fast and precisely it can move, and how efficiently it can operate. This raises requirements for higher torque density, better thermal management, lower weight and greater energy efficiency. Sensing will also become increasingly important as humanoids gain greater autonomy and take on more complex tasks. We therefore do not see the component market as mature or structurally fixed; continued innovation in component design and technical solutions should continue to evolve the competitive landscape.

- **Component redesign remains an important path to higher performance:** In motors, suppliers are adopting higher-grade magnets and higher slot-fill factors to improve torque density and reduce heat generation. In reducers, Laifual is developing a dual-rigid-gear harmonic reducer that integrates the bearing and rigid gear into a single structural component. This design could enable a more compact and lightweight reducer while improving torque density and stiffness.
- **Richer sensing is required to further expand functional capabilities:** Visual-tactile sensors, for example, have attracted increasing attention. Unlike traditional capacitive or piezoresistive solutions, visual-tactile sensing uses cameras to capture deformation in a gel or silicone pad embedded in the fingertip. In theory, this enables higher-resolution, multimodal signal capture while leveraging advances in computer vision. However, costs remain high and several technical barriers have yet to be resolved, including durability, size and interoperability with existing algorithms. Initial adoption is therefore emerging primarily in higher-end dexterous hands.
- **Linear actuator adoption remains limited, but could increase as the technology matures:** Most China humanoids already in production, such as Unitree, UBTECH and Galbot, rely primarily on rotary actuators. Although screw suppliers such as [Hengli Hydraulic](#) have entered humanoid programs at several companies, including Xiaomi, Xpeng, and Li Auto, progress and volumes remain relatively small. Adoption is limited by technical complexity, costs and capacity. For the longer

term, adoption could rise if leading platforms demonstrate clear advantages in payload, efficiency, lifespan and precision.

- **Materials innovation improving weight-performance trade off:** A leading China integrator's new model uses magnesium alloy for the main structure with titanium reinforcement at high-stress points, replacing traditional aluminum.

Exhibit 7: Humanoid diagram



Source: Morgan Stanley Research

Brain/model: From VLA toward world models, but different architectures are likely to co-exist

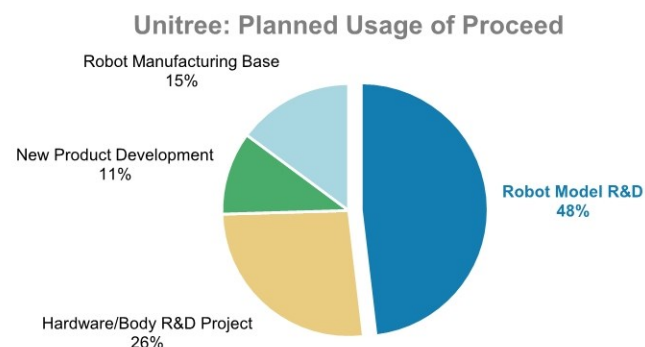
Robot AI architectures are evolving rapidly, with the industry increasingly shifting its attention from conventional VLA models toward world models and world-action models, which add predictive, model-based reasoning. Almost all leading Chinese players we spoke with now claim to be developing proprietary world-model capabilities, suggesting that world models are becoming the industry's next major R&D focus.

But we think multiple paths could co-exist. World models still face meaningful challenges in achieving the generalization, inference latency, efficiency and reliability required for real-world deployment. By contrast, VLA models combined with real-world reinforcement learning (RWRL) could be better suited to executing defined tasks with higher efficiency and predictability. We therefore expect the architecture to evolve toward a hybrid, or dual-system, approach: world models providing higher-level prediction, reasoning and planning, while VLA/RWRL models handle efficient and reliable task execution.

For humanoid integrators, the more durable moat could be the ability to tightly integrate hardware and software, with the data pipeline continuously converting deployment experience into better models. The competitive advantage should increasingly shift toward companies capable of building a scalable deployment-data-model flywheel.

Exhibit 8: The model path is constantly shifting

Source: Unitree, Wang Xingxing, Morgan Stanley Research.

Exhibit 9: Unitree plans to invest nearly half of its IPO proceeds toward "brain" R&D

Source: Unitree, Morgan Stanley Research

Data: Moving toward a scalable data pyramid

Data remains one of the key bottlenecks for embodied intelligence. Just as text is the fuel for LLM training, robot foundation models require large-scale, high-quality data covering physical interactions. The industry is moving toward a data pyramid, combining different sources with varying levels of scalability and embodiment specificity:

- **Robot teleoperation** remains valuable for precise embodiment-specific actions.
- **UMI and portable human interfaces** can collect natural manipulation demonstrations without requiring a robot for every sample.
- **Ego/first-person video** provides large-scale information about tasks, objects and human-world interaction.
- **Simulation** can generate rare conditions and failures cheaply.

Basic data-collection capacity is becoming an ecosystem-wide effort and could commoditize quickly. Since last year, central and local governments in China have established embodied-intelligence data-collection centers, while a growing number of startups are providing outsourced collection, annotation and processing services. At [WAIC 2026](#), we observed relatively limited differentiation across many basic collection offerings, with continued price deflation. However, we do not think all data will commoditize equally. Value is likely to migrate from raw collection hours toward data quality, task design, multimodal synchronization, data cleaning and selection, and model-linked evaluation.

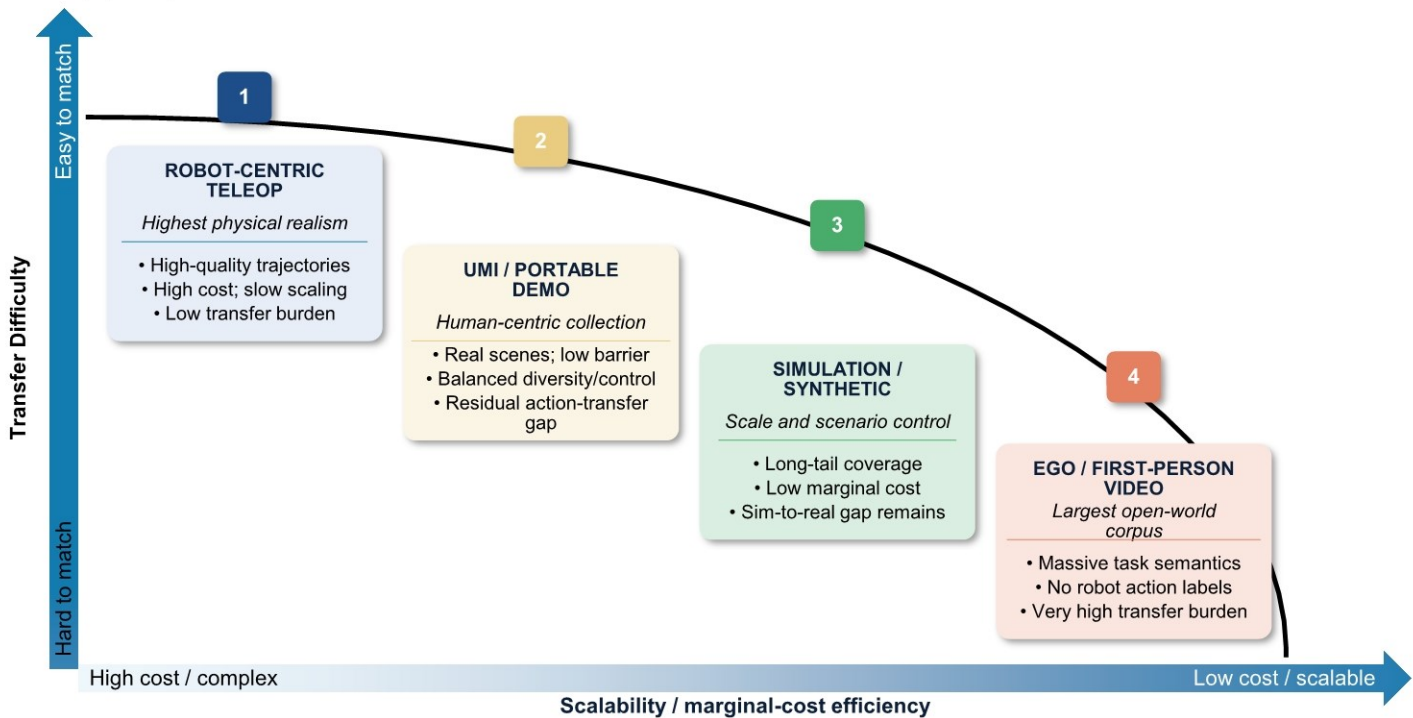
The more important question is whether larger and cheaper datasets translate into better robot capability, i.e. the embodied scaling law. Several company participants we spoke with have already accumulated hundreds of thousands to millions of hours of training data, with some targeting another ~10x increase by year-end or 2027. What matters is whether additional pre-training data improves generalization and reduces the amount of robot-specific data required for new tasks.

- **Dyna-2 provides early evidence of human-to-robot transfer scaling law.** In August 2026, [Dyna-2 World Action Model](#) reported that increasing human-video pre-training data from 1k to 10k, 100k and 1mn hours produced meaningful

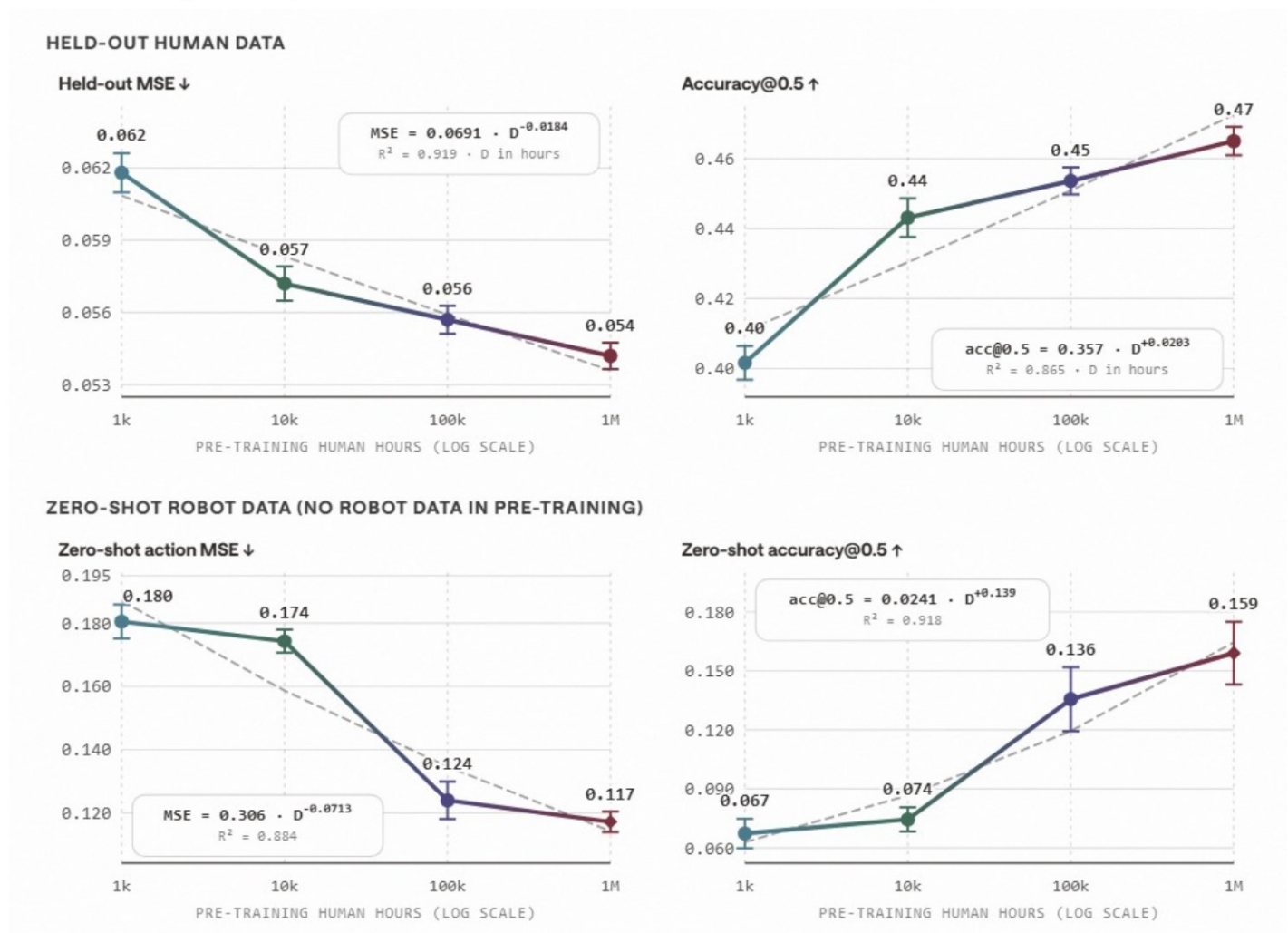
correlated improvements. This suggests that sufficient large-scale human video can contribute to cross-embodiment learning, potentially allowing embodied models to scale without requiring robot-native data to grow at the same rate.

Over time, the stronger moat may come from deployment. For now, real operating data remains a relatively small share of most embodied-AI training datasets. But as deployments ramp up, robots operating at customer sites generate failures, human interventions and successful recoveries that are difficult to reproduce in controlled environments. Fleet size alone is unlikely to be enough; the value of deployment data will depend on operating-environment diversity, edge-case density, data rights, feedback-loop speed and the ability to turn new data into measurable model improvements.

Exhibit 10: An overview of the embodied data landscape for training models, mapped across Transfer Difficulty (Y-axis) and Scaling Difficulty (X-axis).



Source: Siyin Wang et. al. - World Action Models: The Next Frontier in Embodied AI, Morgan Stanley Research.

Exhibit 11: Scaling laws for Dyna-2 world-action models trained on a ladder of human video data, from 1k hours to 1mn hours

Source: Dyna Robotics - "Dyna-2: A 1-Million-Hour Scaling Law for World-Action Models".

Applications: Where are we in the deployment curve?

Deployment remains early, but is increasingly moving toward real-world tasks: In 9M25, 74% of Unitree's humanoid revenue came from R&D and education, versus 17%/9% from commercial & consumer/industrial applications ([Exhibit 12](#)). In 1H26, there was an increasing focus on task-oriented deployment, particularly in industrial applications, reception and guiding ([Exhibit 2](#)). We acknowledge that solutions are still developing and will take time to scale. The market also remains fragmented, given limited generalization capabilities and the still engineer-intensive nature of deployment.

Applications are broadening: Current deployments remain concentrated in relatively structured pick-and-place tasks, including material/box handling, logistics sorting and utility inspection. However, companies are increasingly exploring more complex workflows. Recent examples include:

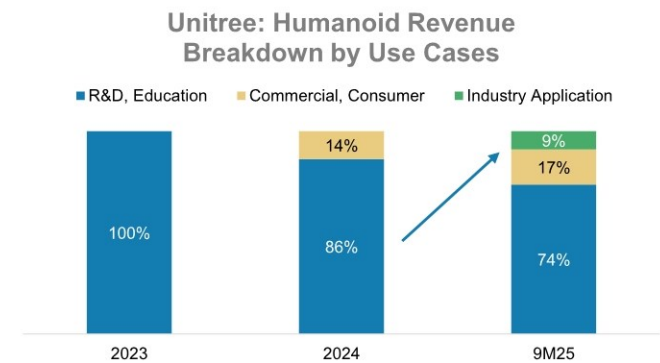
- **PsiBot and YOFC** reportedly launched an embodied-AI solution covering the full workflow from optical-module insertion and inspection to quality sorting and packaging.

- UBTECH and Hitachi have moved Walker S2 into on-site validation at [Hitachi Elevator manufacturing facilities](#), with secondary development being carried out around actual manufacturing processes.
- UBTECH is also partnering with [Terra Robotics](#), a European robotics integrators that has purchased several dozen full-size humanoid, with robots entering a long-term pilot in ROSSMANN logistics facilities.
- [State Grid](#) has placed a Rmb6.8bn (~US\$940mn) order to procure 500 humanoid robots, 3k dual-arm robots and 5k quadrupeds.

We expect rapid application verification to continue through 2026-27. Deployment should gradually move from single tasks in structured environments toward multi-step workflows in semi-structured environments as reliability improves, data accumulates and models become more capable. Household applications, however, face much higher requirements for generalization, safety and reliability; we therefore expect deployment volumes to remain limited in the near term.

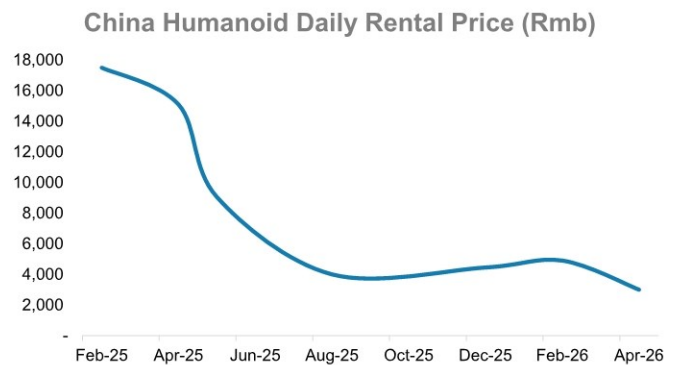
The decline in humanoid rental prices since early 2025 also suggests that novelty- and entertainment-driven monetization is normalizing. In our view, this underscores that the next phase of value creation needs to come from productive work rather than demonstrations.

Exhibit 12: In 9M25, Unitree revenue was still heavily skewed toward R&D and education



Source: Unitree, Morgan Stanley Research.

Exhibit 13: Daily rental prices have been falling since the peak in Feb 2025 (Unitree's first appearance in CCTV new year gala); demand should shift from entertainment to real world work



Source: Juzzyear, Morgan Stanley Research.

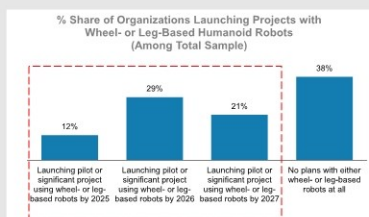
Where will demand come from? Our AlphaWise survey showed adoption willingness is high, while product readiness is a constraint

Our first [AlphaWise survey](#) last year indicated a high willingness to adopt, with products to be improved: 62% of respondents are likely to adopt over 2025-27, indicating significant potential. However, products are not ready; only 23% of respondents are satisfied with current products. Dexterity, functionality and price are key areas to improve. Therefore, enterprises increasingly understand where humanoids could be used; what they are waiting for is sufficient reliability, task success rate and economics to justify replacing or supplementing human labor.

Where is the demand coming from?

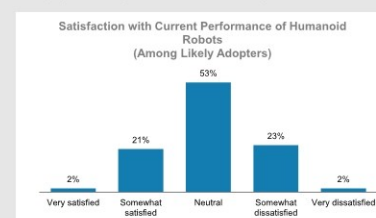
- **Manufacturing and logistics optimization** is a dominant theme, aiming for fully automated systems. Enterprises plan to deploy robots to streamline production lines and logistics for tasks such as assembly, packaging, quality inspection and transport.
- **Customer service and facility management roles** are gaining attention, serving as intelligent guides in real estate offices, performing cleaning and delivery tasks in property management and assisting with public-facing roles such as in-hall guidance and customer service.
- **High-risk and repetitive tasks** remain a top priority, with robots targeted for (1) manufacturing, (2) hazardous maintenance, (3) patrol and inspections, (4) material handling in construction, and (5) underwater operations. These applications aim to improve safety and reduce manual labor in dangerous environments.
- **Broader use case demand spans industries such as construction and agriculture**, which now receive less attention but have huge labor replacement potential considering China employment composition (farming, forestry, fishery accounted for 21% of total employment in China, with construction at 11%).

Exhibit 14: 62% are 'likely adopters' with plans to launch projects using wheel-based or leg-based humanoid robots over the next 3 years



Source: AlphaWise, Morgan Stanley Research. Total respondents n=86. For details, please see [Humanoids: China Humanoids Survey: High Willingness to Adopt Meets Premature Products \(2 Dec 2025\)](#).

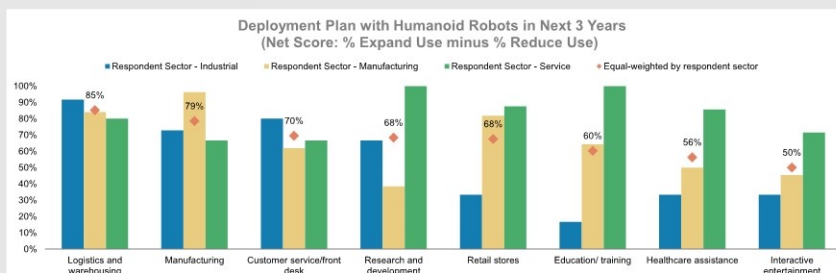
Exhibit 15: Satisfaction with current humanoid robots is mixed, suggesting room for improvement



Source: AlphaWise, Morgan Stanley Research. Likely adopters n=53. For details, please see [Humanoids: China Humanoids Survey: High Willingness to Adopt Meets Premature Products \(2 Dec 2025\)](#).

Exhibit 16:

Logistics/manufacturing/customer services are more likely to utilize humanoid robots over the next 3 years



Source: AlphaWise, Morgan Stanley Research. Likely adopters n=53. For details, please see [Humanoids: China Survey \(2\): What work are humanoids expected to do? \(14 Dec 2025\)](#).

Appendix

Policy overview

Exhibit 17: China's combination of top-down direction, local funding, R&D and application subsidies, real-world training sites and SOE participation is unusually comprehensive...

National Policies	Release Date	Target
MIIT Guiding Opinions On The Innovative Development Of Humanoid Robots	Jan-23	By 2025, products reach internationally advanced levels and achieve mass production; cultivate 2–3 ecosystem companies with global influence
Standards System For Humanoid Robots And Embodied Intelligence	Feb-26	Covers six pillars: basic commonality, brain/intelligent computing, limbs and components, robots and systems, applications, and safety and ethics
“15th Five-year Plan”	Mar-26	Lists robotics as a strategic emerging industry for the first time
2026 Humanoid Robots And Embodied Intelligence Real-world Training Special Initiative	Jun-26	- By end-2026, humanoid robots complete application validation and routine deployment across a set of representative scenarios - Forming 100+ high-value applications, targeting 10k-level deployment capacity - Identifying real operational sites (≥20 per province, ≥10 per central SOE)
Local Government Policies	Release Date	2027 Target
Zhejiang Province Implementation Plan For Innovative Development Of The Humanoid Robot Industry (2024–2027)	Sep-24	- Provincial annual robot output reaches 20k units - Reaching core-industry scale of Rmb20bn; related-industry scale of Rmb50bn
Shenzhen Embodied Intelligence Robot Technology Innovation And Industrial Development Action Plan (2025–2027)	Mar-25	- Cultivating 10+ new companies valued over Rmb10bn, 20+ companies with revenue over Rmb1bn - Achieving 50+ billion-yuan-level application deployments; related-industry scale of >Rmb100bn
Shanghai Embodied Intelligence Industry Development Implementation Plan	Jul-25	- Reaching industry scale of Rmb50bn - Providing 30% subsidy for model R&D, 20% for application projects, 50% for ecosystem projects
Beijing Economic-technological Development Zone: Measures To Promote Innovative Development Of Embodied Intelligence Robots	Aug-25	- Forming 10k-unit-scale mass-production capability for embodied intelligence robots - Building tens-of-millions-scale real-world embodied dataset
Hangzhou Embodied Intelligence Robot “Chain Strengthening” Action Plan (2026–2027)	Jan-26	- Integrator enterprises’ total output exceeds Rmb20bn - Value-chain total output exceeds Rmb50bn

Note: Includes all notable funding disclosed based on our knowledge and research, not all inclusive. Source: Government announcements, Morgan Stanley Research.

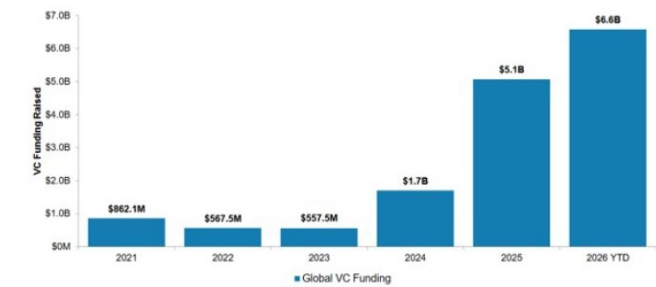
Exhibit 18: Other countries are also introducing increasingly ambitious robotics policies

Country	Date / status	Major policy / initiative	Key details	Targets
South Korea	Apr-25	K-Humanoid Alliance	MOTIE launched a dedicated humanoid alliance with ~40 companies, universities and other organizations.	- Robot-AI foundation model by 2028. - By 2028 humanoid hardware targeted at ≤60 kg weight, ≥50 DoF, ≥20 kg payload, ≥2.5 m/s speed.
	Feb-26	K-Humanoid Moonshot	The government designated humanoids as a core K-Moonshot national mission and established a strategic council. Launched the Public-Private AI Humanoid Core Technology Advancement Project (2026–2030) in May, to invest KRW50.4bn through 2030.	Korea aims to establish humanoid production capacity of 30,000 units annually, by 2030.
Japan	Jun-26	AI Robotics Strategy	Major shift toward Physical AI , domestic multimodal robotics foundation models, robot adoption and large-scale social implementation. Noetra + AIST were selected to lead a domestic multimodal foundation-model project.	Japan aims 10m robots (not only humanoids) deployed by 2040 across 18 sectors, including manufacturing, food, healthcare etc.
US	Jul-26	FCC included foreign-made advanced robotic devices into the Covered List	The addition prohibits FCC authorization, generally blocking importation, marketing, or sale in the US . Robots that had already received FCC authorization are not affected.	-
UK	Jan-26	Robotics Adoption Programme + Regulatory Innovation Office robotics priority	Focus is on accelerating commercial adoption of broad robotics and removing regulatory barriers.	~5 Robotics Adoption Hubs, expected to operate from 2H26. Each receives at least £2m over four years.

Source: Government announcements and news reports, Morgan Stanley Research.

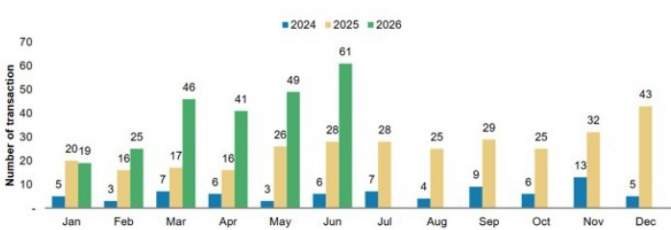
Financing overview

Exhibit 19: 2026 global venture funding for humanoid robots has already exceeded the sum of 2025



Source: As of 7/15/2026. Includes funding for companies classified under 'Humanoid Robotics' in PitchBook Source: PitchBook, Morgan Stanley Research

Exhibit 20: China humanoids financing activities continued to accelerate



Note: Includes all notable funding disclosed based on our knowledge and research. However, there may still be some not captured. We updated number of transaction in previous months with transaction announced after our published previous Humanoids Horizon. Source: Company announcements, GGII, Morgan Stanley Research.

Exhibit 21:

China is seeing broader, faster, and larger size in humanoid companies' financing

Date	Country	Company	Round	Fund raised (reported currency)	Fund raised (US\$'mn)	Valuation
Jan-26	US	Skild AI	Series C	US\$1.4bn	1,400	>US\$14bn
Feb-26	China	LimX Dynamics	Series B	US\$200m	200	-
Feb-26	China	Beijing Innovation Center of Humanoid Robotics	Series A	>Rmb700m	100	-
Feb-26	US	Appronik	Series A extension	US\$520m	520	~US\$5bn
Feb-26	China	Galaxea AI	Series B	Rmb1.0bn	148	~Rmb10bn (~US\$1.5bn)
Feb-26	China	AI² Robotics	Series B series	>Rmb1.0bn	148	>Rmb10bn (~US\$1.5bn)
Mar-26	China	Galbot	Strategic financing	Rmb2.5bn	362	-
Mar-26	China	Noetix Robotics	Series B	~Rmb1.0bn	148	-
Mar-26	China	GigaAI	Pre-B	~Rmb1.0bn	148	-
Mar-26	China	PaXini	Series B	>Rmb1.0bn	148	>Rmb10bn (~US\$1.5bn)
Mar-26	China	PsiBot	Angel + Pre-A	~Rmb2.0bn	296	-
Mar-26	US	Rhoda AI	Series A	US\$450m	450	US\$1.7bn
Mar-26	China	LightWheel	Series A++ / A+++	Rmb1.0bn	148	-
Mar-26	US	Sunday Robotics	Series B	US\$165m	165	US\$1.15bn
Mar-26	US	Mind Robotics	Series A	US\$500m	500	~US\$2.0bn
Mar-26	China	D-Robotics	Series B1	US\$120m	120	-
Mar-26	China	Robotera	Strategic round	Rmb1.0bn	148	>Rmb10bn (~US\$1.5bn)
Apr-26	China	Galaxea AI	Series B+ / follow-on	Rmb2.0bn	291	>Rmb20bn (~US\$3.0bn)
Apr-26	China	Spirit AI	Follow-on	Rmb1.0bn	148	>Rmb10bn (~US\$1.5bn)
Apr-26	China	EngineAI	Series B	~US\$200m	200	>Rmb10bn (~US\$1.4bn)
Apr-26	China	TARS Robotics	Pre-A	US\$455m	455	-
May-26	China	ROBOTERA	Strategic follow-on	>US\$200m	200	-
Jun-26	China	Coowa Robot	New financing	>US\$600m	600	>US\$3bn
Jun-26	China	AI² Robotics	New / growth financing	~Rmb5.0bn	735	>Rmb20bn (~US\$3.0bn)
Jul-26	China	Morphi	Angel series	>Rmb1.0bn	148	>Rmb7.0bn (~US\$1bn)
Jul-26	China	LimX Dynamics	Pre-IPO	~US\$200m	200	-
Jul-26	US	Walden Robotics	Seed	US\$300m	300	US\$1.1bn
Jul-26	UK	Humanoid	Series A	US\$152m	152	US\$1.35bn
Jul-26	US	Atoms	Venture financing	US\$1.7bn	1,700	-

Source: Company and news announcements, Morgan Stanley Research.

Valuation Methodology and Risks

Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ)

We apply a 25x 2026e P/E for the core business, which we view as appropriate to use 25x (5-year average FY1 P/E) considering the sustained growth across business lines. In terms of humanoid potential, we use 6x P/S on Rmb1.1bn sales when global humanoid sales exceeded 1mn units to derive a humanoid business value, which is justified by a longer application timeline for new reducers but higher growth potential.

Risks to Upside

- Increasing penetration of dual/tri-motor and coaxial gearboxes
- Faster capacity expansion overseas, capturing more overseas OEMs' demand
- Faster-than-expected intelligent actuator growth driven by diversifying downstream
- Faster-than-expected progress of humanoid reducers

Risks to Downside

- Slower-than-expected share gains
- Weaker-than-expected overseas market demand
- Intensifying gear and actuator competition in China

Jiangsu Hengli Hydraulic Co.Ltd (601100.SS)

Non-humanoid business: Derived from a target 2027 P/E of 30x. This is lower than last upcycle's peak multiple of ~40-50x, as we believe: 1) this upcycle's less-robust domestic peak sales should mean lower peak multiple; 2) its adjust net profit growth is more normalized; and 3) its Pump & Valve business has past the ultra-high growth stage.

Humanoid business: Derived from a DCF model with a WACC of 11% and a terminal growth rate of 4%.

Risks to Upside

- Stronger-than-expected excavator and pumps & valves demand.
- Notable penetration into foreign brands' supply chain.
- Faster-than-expected humanoid robots penetration and market share gain in the supply chain.

Risks to Downside

- Sharp decline in excavator and pumps & valves demand in China.
- Failure to expand its share in non-excavator parts.
- Slower-than-expected humanoid robot penetration.

Leader Harmonious Drive Systems (688017.SS)

Base case. We use DCF to value Leaderdrive, by discounting 2025-50e cash flow at a WACC of 11% and terminal growth rate of 4%, to derive per share value of Rmb631.

Risks to Upside

- Faster growth in industrial robot demand
- Faster humanoid adoption
- Penetrating into more integrators' supply chain
- Margin expansion with high utilization and favorable product mix

Risks to Downside

- Slower humanoid volume growth
- Lower penetration of harmonic reducers in humanoids
- Market-share erosion and failure to enter leading US integrators' supply chain
- Intensified competition, greater pricing pressure

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(as of July 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Stock Price, Price Target and Rating History (See Rating Definitions)

Jiangsu Hengli Hydraulic Co.Ltd (601100.SS) - As of 08/17/26 GMT in CNY
Industry : China Industrials



Stock Rating History: 8/1/21 : E/I; 2/4/23 : E/A; 5/23/23 : O/A; 1/8/25 : O/I

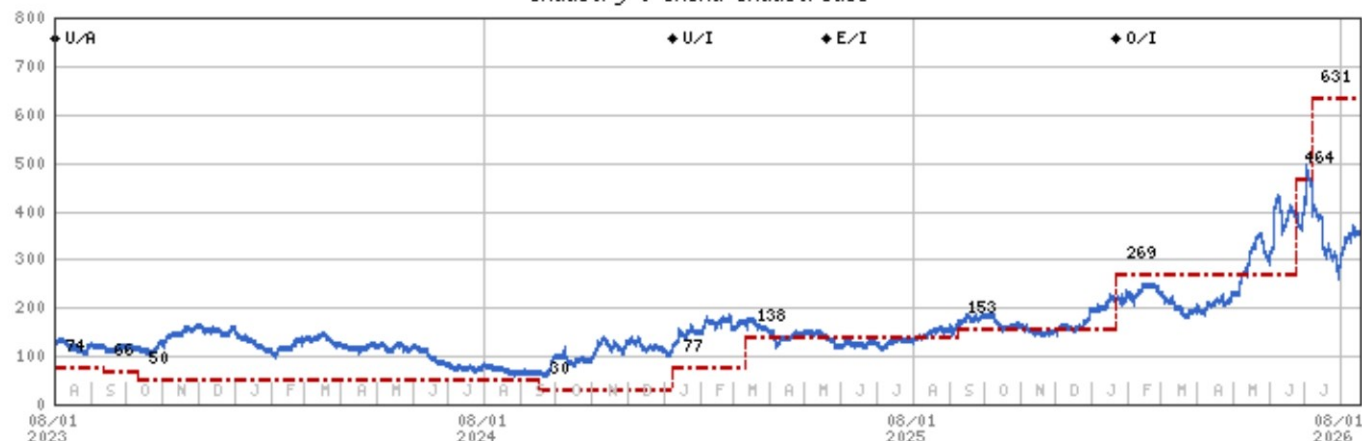
Price Target History: 3/17/21 : 95; 8/16/21 : 90; 8/24/21 : 83; 6/30/22 : 59; 9/29/22 : 52; 12/13/22 : 55; 5/23/23 : 63;
7/11/23 : 68; 10/11/23 : 71; 1/3/24 : 68; 2/26/24 : 67; 4/25/24 : 61; 8/28/24 : 63; 10/8/24 : 69; 11/5/24 : 63; 3/12/25 : 104;
5/19/25 : 99; 7/14/25 : 93; 9/8/25 : 105; 1/22/26 : 133; 8/11/26 : 150

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Leader Harmonious Drive Systems (688017.SS) - As of 08/17/26 GMT in CNY
Industry : China Industrials



Stock Rating History: 8/1/21 : O/I; 6/30/22 : E/I; 2/4/23 : E/A; 5/23/23 : U/A; 1/8/25 : U/I; 5/19/25 : E/I; 1/22/26 : O/I

Price Target History: 4/28/21 : 99.33; 8/20/21 : 112.09; 1/10/22 : 142.29; 6/30/22 : 95.71; 8/15/22 : 140; 10/31/22 : 107;
12/13/22 : 91; 5/23/23 : 74; 9/11/23 : 66; 10/11/23 : 50; 9/16/24 : 30; 1/8/25 : 77; 3/12/25 : 138; 9/8/25 : 153; 1/22/26 : 269;
6/23/26 : 464; 7/7/26 : 631

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Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
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Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ) - As of 08/17/26 GMT in CNY
Industry : China Industrials



Stock Rating History: 8/25/23 : 0/A; 1/6/25 : 0/I

Price Target History: 8/25/23 : 39; 1/5/24 : 30; 2/27/24 : 29; 1/6/25 : 37; 3/12/25 : 44; 5/19/25 : 43; 9/8/25 : 44; 1/22/26 : 50

Source: Morgan Stanley Research

Date Format: MM/DD/YY

Price Target --

No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ■

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: China Industrials

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/14/2026)
Chelsea Wang		
China Railway Group (601390.SS)	E (05/12/2022)	Rmb4.36
China Railway Group (0390.HK)	E (08/11/2025)	HK\$3.40
China State Construction Engineering (601668.SS)	U (08/11/2025)	Rmb4.40
Han's Laser (002008.SZ)	O (10/02/2025)	Rmb99.00
Hefei Meyer Optoelectronic Technology (002690.SZ)	E (09/08/2025)	Rmb15.25
iRay Technology Company Limited (688301.SS)	O (07/07/2026)	Rmb97.30
Neway Valve (Suzhou) Co., Ltd (603699.SS)	O (09/12/2025)	Rmb53.33
Shanghai BOCHU Electronic Technology (688188.SS)	O (08/22/2024)	Rmb102.82
Shenzhen Envicool Technology Co Ltd (002837.SZ)	O (08/19/2024)	Rmb57.09
Sheng Zhong		
Beijing Geekplus Technology Co., Ltd. (2590.HK)	O (08/07/2025)	HK\$10.11
Centre Testing International Group (300012.SZ)	E (11/18/2024)	Rmb13.98
CRRC Corp Ltd (1766.HK)	U (01/22/2026)	HK\$4.94
CRRC Corp Ltd (601766.SS)	U (01/22/2026)	Rmb5.88
DR Laser (300776.SZ)	E (12/17/2021)	Rmb121.45
Estun Automation Co Ltd (002747.SZ)	U (06/30/2022)	Rmb34.75
Haitian International Holdings Limited (1882.HK)	E (09/08/2025)	HK\$20.36
Hongfa Technology Co Ltd (600885.SS)	O (05/23/2023)	Rmb36.20
Jiangsu Guomao Reducer Co Ltd (603915.SS)	U (01/08/2025)	Rmb13.26
Jiangsu Hengli Hydraulic Co.Ltd (601100.SS)	O (05/23/2023)	Rmb106.28
Jingsheng Mechanical & Electrical Co (300316.SZ)	U (01/08/2025)	Rmb43.49
Leader Harmonious Drive Systems (688017.SS)	O (01/22/2026)	Rmb349.93
Sany Heavy Industry Co., Ltd. (600031.SS)	O (01/08/2025)	Rmb18.75
Shenzhen Inovance Technology (300124.SZ)	++	Rmb61.20
Shenzhen SC New Energy Technology Corp (300724.SZ)	U (09/08/2025)	Rmb58.60
Sinotruk (Hong Kong) Limited (3808.HK)	E (05/19/2025)	HK\$38.98
Suzhou Maxwell Technologies Co Ltd (300751.SZ)	U (09/15/2023)	Rmb178.39
Times Electric (3898.HK)	E (01/22/2026)	HK\$34.72
WeiChai Power (2338.HK)	O (03/30/2026)	HK\$35.02

WeiChai Power (000338.SZ)	O (03/30/2026)	Rmb29.65
Wuxi Autowell Technology Co Ltd (688516.SS)	U (09/08/2025)	Rmb41.55
Wuxi Lead Intelligent (300450.SZ)	O (09/08/2025)	Rmb34.78
Zhejiang Dingli Machinery Co Ltd. (603338.SS)	O (11/05/2025)	Rmb55.60
Zhejiang Hangke Technology (688006.SS)	O (09/08/2025)	Rmb21.90
Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ)	O (08/25/2023)	Rmb37.89
Zoomlion Heavy Industry (1157.HK)	O (09/08/2025)	HK\$6.87
Zoomlion Heavy Industry (000157.SZ)	O (09/08/2025)	Rmb6.93

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